

Software Engineering

Lecture 5: System Modeling

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Focus Questions

- What is **System Modeling** and why do software engineers build abstract models?
- What are the **4 complementary perspectives** used to model software systems?
- How do **Context Models** define what is inside vs. outside system boundaries?
- How do **Interaction Models** (Use Cases & Sequence Diagrams) capture dynamic behavior?
- How do **Structural Models** (Class Diagrams, Generalization, Aggregation) represent system organization?
- How do **Behavioral Models** (State Diagrams & Activity Diagrams) handle data and events?
- How can **AI & LLMs** assist in automated System Modeling (UML, Sequence, Class Diagrams)?

What is System Modeling?

- **Abstract Representation:**

- System modeling is the process of developing abstract models of a system, with each model presenting a different view or perspective.

- **Graphical Standard (UML):**

- Almost universally represented using graphical notations based on the **Unified Modeling Language (UML)**.

- **Purposes of System Modeling:**

1. **Discussion:** Facilitating communication between engineers, analysts, and customers.
2. **Documentation:** Documenting existing or proposed system architecture.
3. **Implementation:** Generating executable code (Model-Driven Engineering).

4 System Modeling Perspectives

External (Context)

- System Boundaries
 - Environment
- External Systems

Interaction

- User-to-System
- Component-to-Component
- Use Case / Seq.

Structural

- Class Diagrams
 - Data Models
- Associations & Aggregation

Behavioral

- Dynamic Response
 - Events & Stimuli
- State Diagrams

5 Essential UML Diagram Types

1. **Activity Diagram:** Shows the flow of activities involved in a process or data transformation.
2. **Use Case Diagram:** Shows interactions between a system and its external actors.
3. **Sequence Diagram:** Shows chronological interactions between actors and system objects over time.
4. **Class Diagram:** Shows object classes in the system and associations between them.
5. **State Diagram:** Shows how the system reacts to internal and external events.

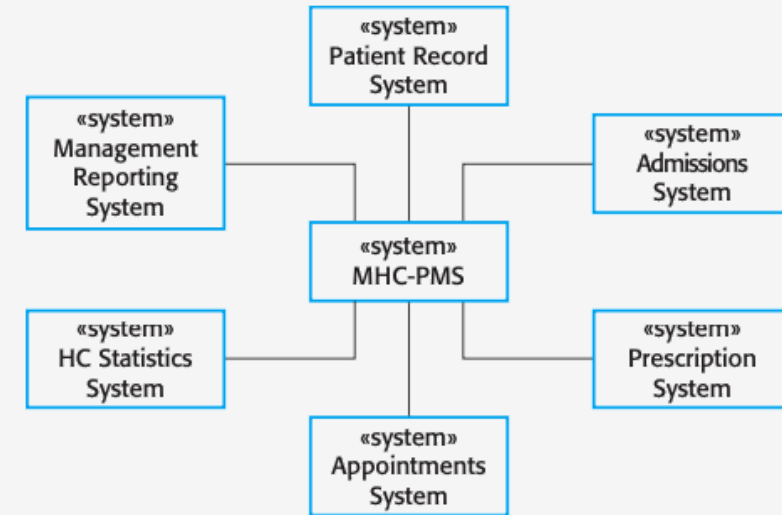
Context Models & System Boundaries

- **Operational Context:**

- Illustrates what lies *outside* system boundaries (other systems, databases, external services).

- **System Boundaries:**

- Defines what is inside vs. outside system scope.
- Dictates system requirements and operational dependencies.



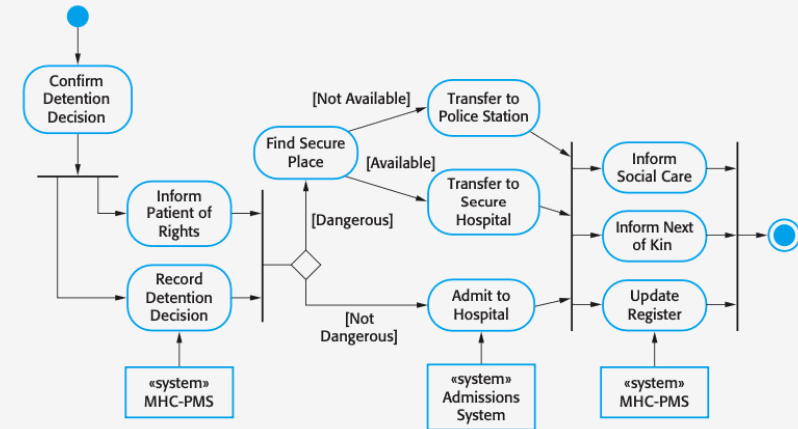
Process Models & Activity Diagrams

- **Business Process Workflow:**

- While context models show system environment, process models show how the system is used in broader business workflows.

- **UML Activity Diagrams:**

- Model sequential, parallel, and conditional activities using swimlanes and decision nodes.



Interaction Models

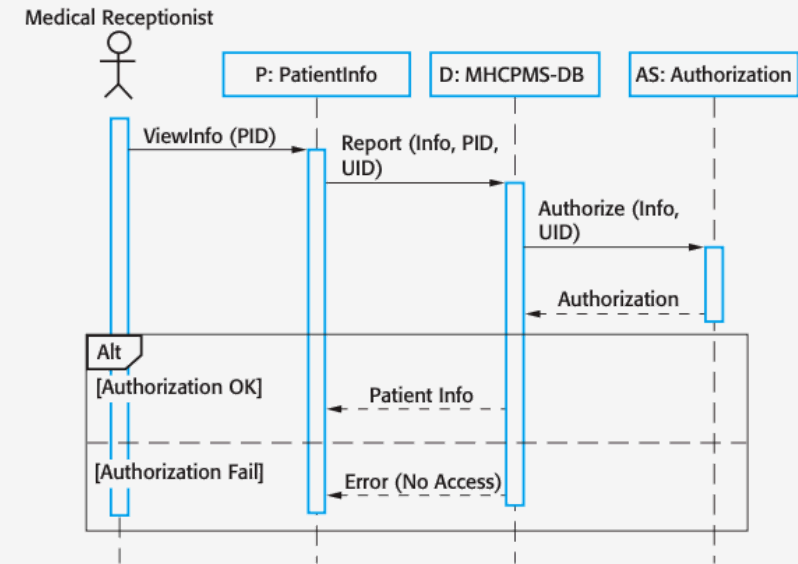
- **Purpose:**
 - Models interactions between **users and system**, **system to external systems**, or **internal components**.
- **Key UML Diagrams:**
 - **Use Case Diagrams:** High-level summary of system functions and external actors.
 - **Sequence Diagrams:** Detailed chronological message exchange between objects.

Use Case Relationships: Include vs. Extend

- **<<include>>** (Mandatory Relationship):
 - The base use case *always* incorporates the behavior of the included use case.
 - Used to factor out common shared functionality (e.g., *Withdraw Cash* **<<include>>** *Authenticate User*).
- **<<extend>>** (Optional / Conditional Relationship):
 - The base use case is extended by another use case *only under specific conditions*.
 - Adds optional behavior without modifying the base use case (e.g., *Place Order* **<<extend>>** *Apply Coupon*).

Sequence Diagrams

- **Chronological Message Flow:**
 - Objects and actors are listed along the top; vertical lifelines represent time progression.
- **Annotated Arrows:**
 - Synchronous calls, asynchronous messages, and return values detail object interactions during a use case instance.



Concept Check Question 1

Which UML diagram is specifically designed to show chronological message exchanges between actors and system objects over time during a use case execution?

- **A.** Activity Diagram
- **B.** Class Diagram
- **C.** Sequence Diagram
- **D.** State Diagram



Concept Check Question 1: Answer

Which UML diagram is specifically designed to show chronological message exchanges between actors and system objects over time during a use case execution?

- **Correct Answer: C**
- **Explanation:** Sequence diagrams list objects along the top and show chronological message calls down vertical lifelines.



Structural Models & Class Diagrams

- **Static Architecture:**

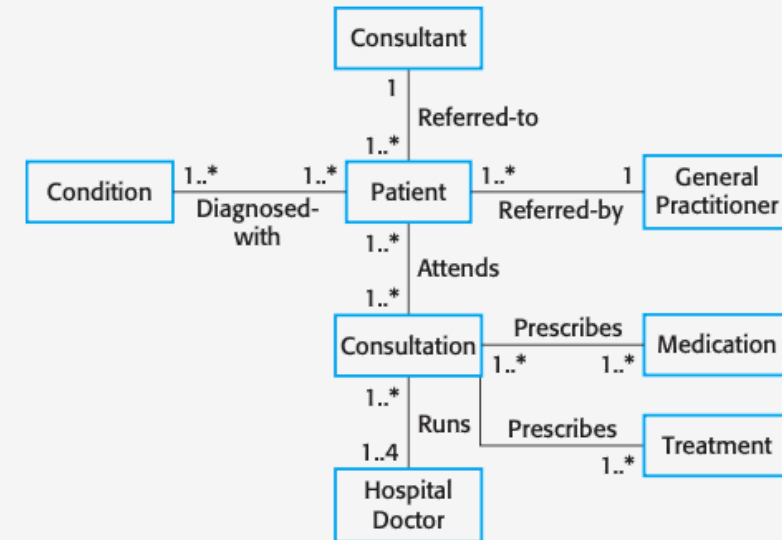
- Displays the organization of a system in terms of object classes and associations.

- **Class Components:**

- Attributes (data) and Operations (methods).

- **Associations:**

- Define links and multiplicity (1:1, 1:N, N:M) between classes.



Generalization (Inheritance)

- **Managing Complexity Through Abstraction:**
 - Higher-level superclasses define common attributes and operations; lower-level subclasses **inherit** these properties and add specific details.
- **Example Hierarchy:**
 - `Doctor` (Superclass) → `HospitalDoctor` and `GeneralPractitioner` (Subclasses) → `Consultant` and `TraineeDoctor`.
- **Object-Oriented Implementation:**
 - Implemented using class inheritance mechanisms (e.g., `extends` in Java/TypeScript).

Aggregation (Whole-Part Relationships)

- **Aggregation Association:**

- Indicates that an object is composed of or contains other object classes.
- Represents a "**part-of**" relationship in semantic data models.

- **Encapsulation & Composition:**

- Helps reduce complexity by grouping related components into higher-level composite structures.

Behavioral Models

- **Dynamic Execution Modeling:**

- Models system behavior when responding to environmental stimuli during execution.

- **Two Types of Stimuli:**

1. **Data-Driven Processing:** System controlled by incoming data streams (e.g., business transaction processing).
2. **Event-Driven Processing:** System driven by external/internal events causing state transitions (e.g., real-time control systems).

State Machine Models

- **Statechart Diagrams:**

- Model system states as nodes and events/stimuli as transition arcs.

- **Finite State Machine:**

- System exists in a finite set of discrete states (e.g., *Waiting*, *Enabled*, *Cooking*, *Disabled*).

- **Real-Time Systems:**

- Ideal for modeling reactive hardware/software systems like microwave ovens, mobile devices, and automotive ACC.

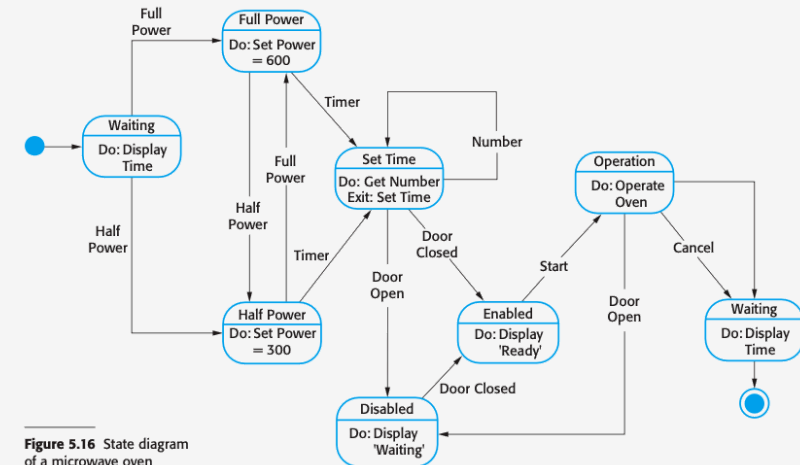


Figure 5.16 State diagram of a microwave oven

AI in System Modeling: Overview

- **The Graphical Modeling Revolution:**

- LLMs convert natural language software specifications into **text-based UML diagrams** (PlantUML, Mermaid.js).
- Serves as an **AI Diagramming Copilot** to rapidly visualize system interactions.

How AI Assists System Modeling (UML & Diagramming Copilot)

1. Text-to-UML

- User Stories to PlantUML / Mermaid
- Sequence Diagram Generation
- **Rapid Visual Prototyping**

2. Domain Model

- Extract Entities from Specs (Nouns)
- Draft Class Attributes/Methods
- **Association Relationships**

3. Consistency

- Cross-Diagram Validation
- Use Case <-> Class Alignment
- **Detect Missing States/Transitions**

4. Reverse Eng

- Code-to-UML Class Diagrams
- Legacy Code Understanding
- **Architect Human Review**

4 Core AI Applications in System Modeling

1. Text-to-UML Generation:

- Generates Sequence, Class, and Activity diagrams directly from user story text.

2. Domain Model Entity Extraction:

- Analyzes SRS documents to extract domain nouns (classes, attributes) and verbs (methods, associations).

3. Cross-Diagram Consistency Validation:

- Scans Use Case actors, Class diagrams, and Sequence lifelines to detect mismatched component names.

4. Code-to-Model Reverse Engineering:

- Analyzes complex legacy codebases to auto-generate UML class structures for developer onboarding.

Human-in-the-Loop: Modeling Risks & Best Practices

- **Risks of Unchecked AI in System Modeling:**
 - **Hallucinated Associations:** Inventing non-existent inheritance or aggregation relationships.
 - **Model Bloat:** Creating over-engineered class hierarchies instead of clean object abstractions.
- **The Golden Rule:**

AI Drafts the Diagram; Engineers Validate the Semantics!

Software Architects must verify that generated UML models accurately represent system domain reality.

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Concept Check Question 2

In UML Use Case modeling, what is the main difference between an `<<include>>` and an `<<extend>>` relationship?

- **A.** `<<include>>` is optional; `<<extend>>` is mandatory.
- **B.** `<<include>>` is mandatory shared behavior; `<<extend>>` is conditional behavior added under specific rules.
- **C.** `<<include>>` applies only to actors; `<<extend>>` applies only to classes.
- **D.** There is no difference in UML.



Concept Check Question 2: Answer

In UML Use Case modeling, what is the main difference between an `<<include>>` and an `<<extend>>` relationship?

- **Correct Answer: B**
- **Explanation:** `<<include>>` represents mandatory shared functionality; `<<extend>>` represents optional/conditional functionality added to a base use case.



Recap: Fill-in-the-blank Quiz

Test your understanding of the core concepts in this chapter:

1. diagrams show the operational boundary between a system and external systems.
2. In UML, a diagram models chronological message passing between objects over time.
3. Class (inheritance) allows subclasses to inherit attributes and operations from a superclass.
4. machine models describe system responses to discrete events by transitioning between states.



Recap: Answers & Summary

Here are the completed concepts:

1. **Context** diagrams show the operational boundary between a system and external systems.
2. In UML, a **sequence** diagram models chronological message passing between objects over time.
3. Class **generalization** (inheritance) allows subclasses to inherit attributes and operations from a superclass.
4. **State** machine models describe system responses to discrete events by transitioning between states.



References

- **Sommerville Software Engineering Book (Chapter 5)**
 - [OMG Unified Modeling Language \(UML\) Specification](#)
 - [Lucidchart UML Modeling Guide](#)